E-COMMERCE 360° ANALYTICS

MySQL + Power BI | College Project Report

An end-to-end study of sales performance, customer intelligence, order fulfilment and payment behaviour

Includes three dashboard views and technical documentation

Academic project | October 2026

1. Executive Summary

E-Commerce 360° Analytics is an end-to-end business intelligence project using a relational MySQL database and a three-page interactive Power BI report. It integrates customer, product, order, order-line and payment data to monitor delivered sales, profitability, customer behaviour, fulfilment and payment preferences. The supplied PBIX contains three report pages; screenshots below document the completed dashboard views.

2. Objectives

Build and validate a structured e-commerce database; model relationships in Power BI; calculate consistent business KPIs with DAX; analyse sales by month, category and geography; understand repeat purchasing and RFM segments; and track order statuses and payment methods.

3. Tools and Workflow

Layer	Technology / role
Data storage	MySQL database: ecommerce_360
Data input	Five UTF-8 CSV tables
Data modelling	Power BI relationships and calculated columns
Business logic	DAX measures and RFM scoring
Presentation	Three-page Power BI dashboard (.pbix)

4. Dataset

Table	Rows	Purpose
Customers	2,000	Customer attributes and geography
Products	100	Product and category reference
Orders	5,000	Order dates, customer links and status
Order_Details	12,536	Order line quantities, sales and profit
Payments	5,000	Order-linked payment method and amount

The project uses a synthetic e-commerce dataset. Values and findings describe this dataset, not an actual retailer.

5. Data Preparation and Validation

CSV files were imported into MySQL and checked for record counts, missing identifiers, duplicates and broken foreign-key references. In Power BI, the five imported tables were assigned suitable data types, including date and decimal fields. Business measures are maintained in a dedicated KPI_Measures table.

6. Relational Model

Relationship	Cardinality / filter direction
Customers[customer_id] → Orders[customer_id]	One-to-many; single
Orders[order_id] → Order_Details[order_id]	One-to-many; single
Products[product_id] → Order_Details[product_id]	One-to-many; single
Orders[order_id] ↔ Payments[order_id]	One-to-one; both

Order line revenue is aggregated through the Orders relationship. Delivered-only revenue and profit are explicitly filtered to order_status = Delivered. This avoids presenting all-status transaction value as fulfilled sales.

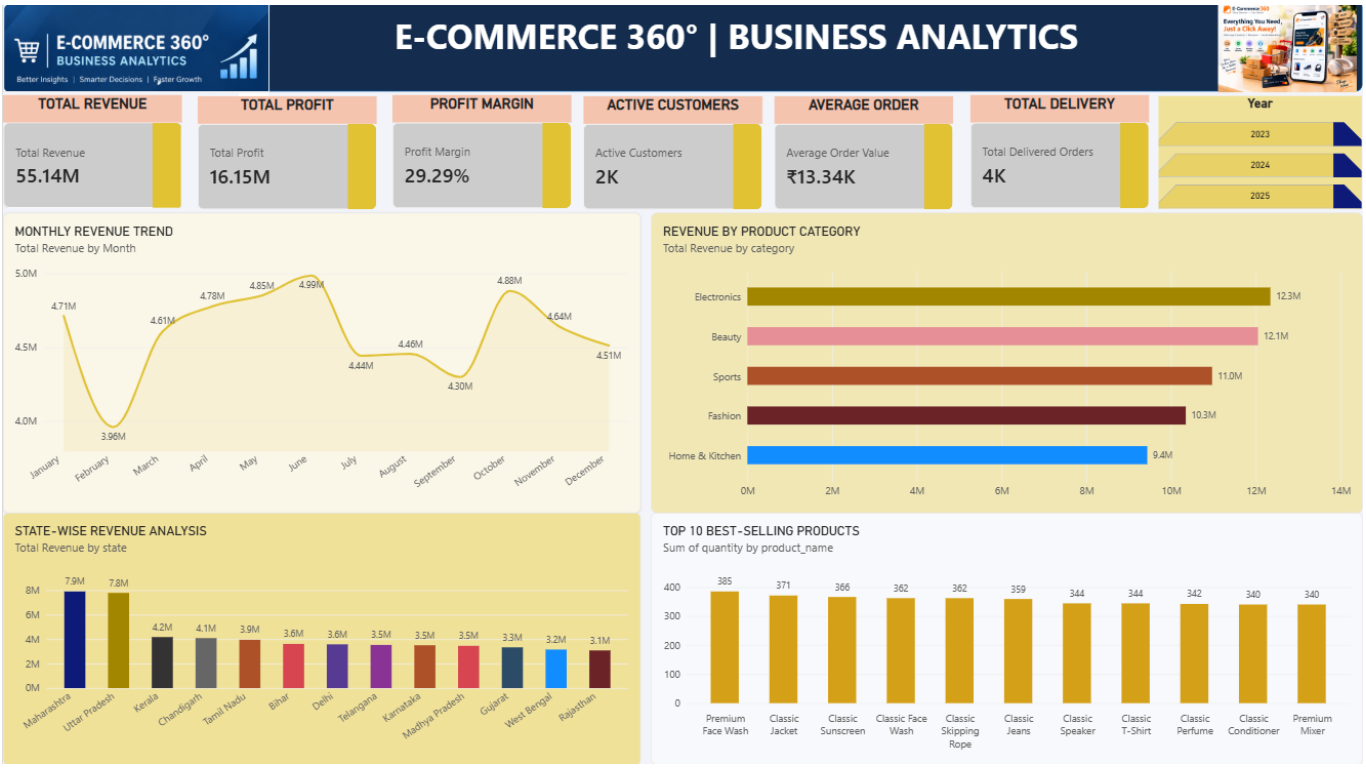
7. Core DAX Measures

Measure	Business definition
Total Orders	DISTINCTCOUNT(Orders[order_id])
Delivered Orders	Distinct order count filtered to Delivered
Total Revenue	SUM(Order_Details[sales_amount]), Delivered only
Total Profit	SUM(Order_Details[profit_amount]), Delivered only
Profit Margin	Total Profit / Total Revenue
Average Order Value	Total Revenue / Delivered Orders
Delivery Success Rate	Delivered Orders / Total Orders
Active Customers	Distinct customers with a Delivered order
Retention Rate	Customers with >1 Delivered order / Active Customers
Payment Orders	DISTINCTCOUNT(Payments[order_id])

Display-only FORMAT measures were used for exact card labels (for example 5,000 instead of 5K). Numeric base measures are retained for calculations and chart aggregation.

8. Dashboard 1 — Executive Sales Overview

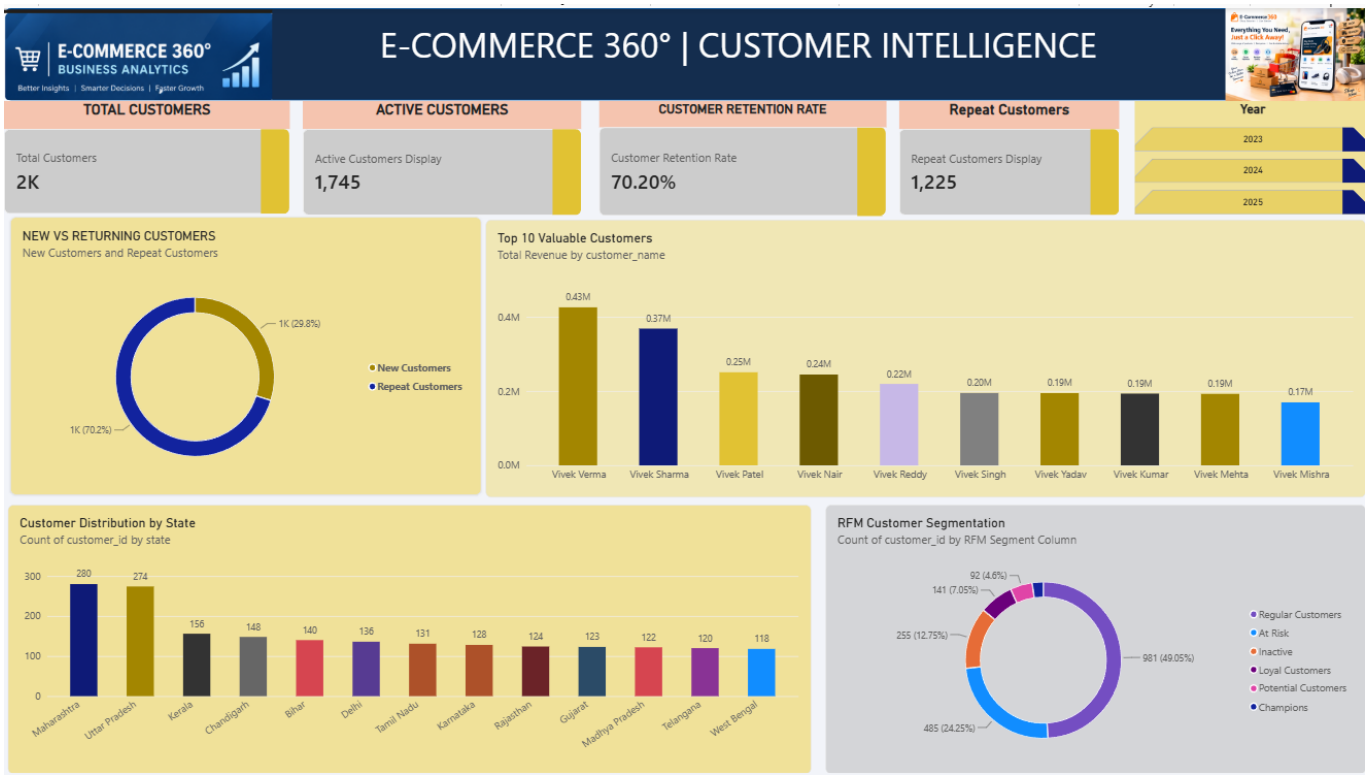
This page combines delivered revenue, profit, margin, active customers, average order value and delivered orders with monthly revenue, category performance, geographic contribution and best-selling products. A Year slicer supports period exploration.



Metric	Observed result
Delivered Revenue	Approximately 55.14 million
Delivered Profit	Approximately 16.15 million
Profit Margin	Approximately 29.29%
Average Order Value	Approximately 13,335.55

9. Dashboard 2 — Customer Intelligence

The customer page examines customer base, active and repeat customers, retention, new-versus-returning composition, top customer value, state distribution and RFM segmentation.

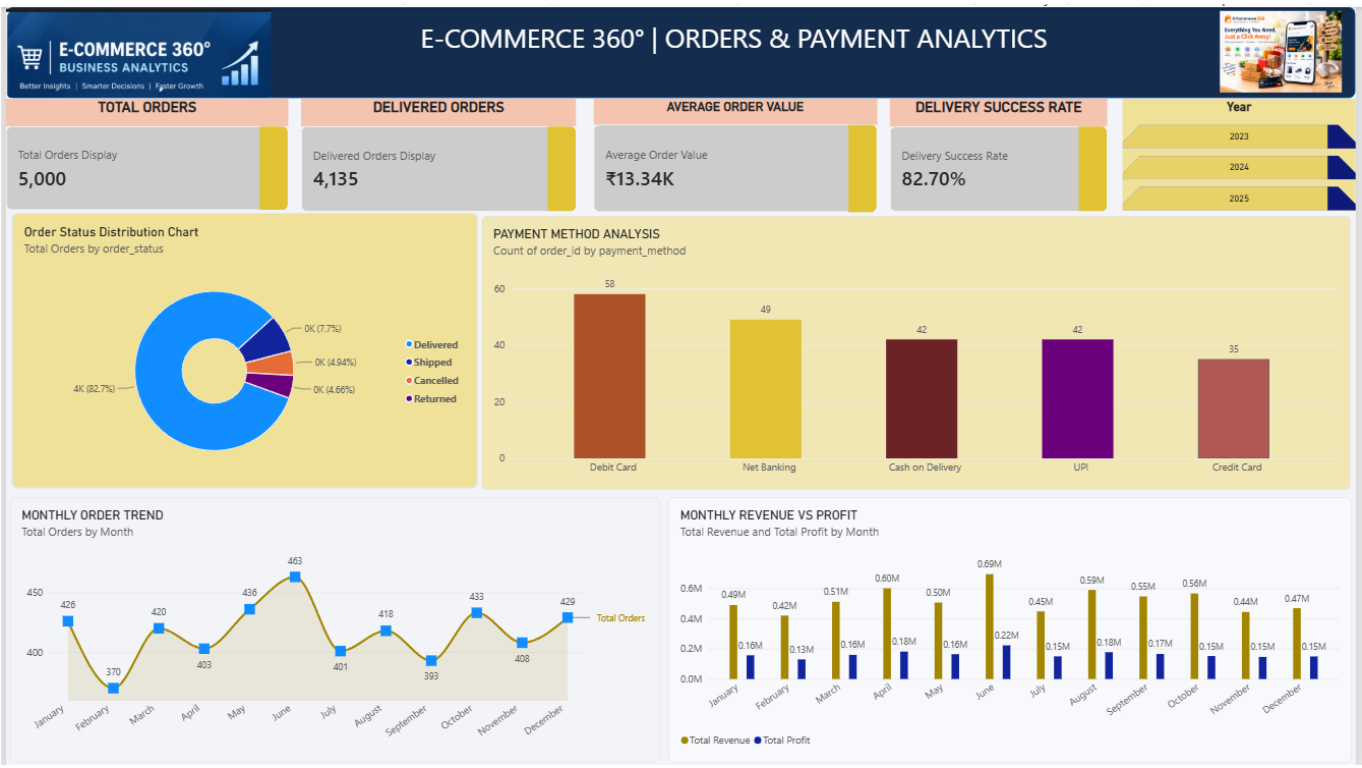


Metric	Observed result
Total Customers	2,000
Active Customers	1,745
Repeat Customers	1,225
New Customers	520
Retention Rate	70.20%

RFM columns use recency, delivered-order frequency and monetary value to classify customers into Champions, Loyal, Potential, At Risk, Regular and Inactive groups. These are snapshot calculated columns and do not dynamically re-score with the Year slicer.

10. Dashboard 3 — Orders & Payment Analytics

The third page combines order volume, delivered orders, AOV and delivery success with order status distribution, payment-method order counts, monthly order trend and monthly revenue-versus-profit comparison.



Metric	Observed result
Total Orders	5,000
Delivered Orders	4,135
Delivery Success Rate	82.70%
Average Order Value	Approximately 13.34K

Note: The screenshot above was captured before the payment-method chart filter was corrected. The Top N filter was subsequently cleared and the Payment Orders measure was applied; the live PBIX is the authoritative final interactive view.

11. Key Findings and Business Insights

Fulfilment: 4,135 of 5,000 orders were delivered, yielding an 82.70% delivery success rate. The remaining statuses warrant separate operational monitoring.

Profitability: Delivered revenue is approximately 55.14 million and profit 16.15 million, corresponding to a margin of roughly 29.29%.

Customer behaviour: 1,225 of 1,745 active customers are repeat customers, giving a 70.20% repeat-customer rate under the project definition.

Merchandising: Category and product rankings help identify sales concentration and guide assortment decisions.

Payment behaviour: Payment-method analysis counts distinct payment-linked orders by method; it is not a chart of payment amount.

12. Interpretation and Limitations

Results depend on the synthetic dataset and selected order-status definitions. Delivered-only KPIs should not be compared directly with unfiltered order or payment counts without a clear label. RFM thresholds are illustrative business rules. Month labels must be chronologically sorted, and year-specific comparisons require a selected year or an appropriate year-month axis.

13. Conclusion

The project demonstrates a complete analytics workflow from relational data preparation through Power BI modelling and interactive storytelling. Its three pages give management complementary views of commercial performance, customer quality, and fulfilment/payment operations.

14. Future Scope

Potential extensions include scheduled refresh, returns/refund analysis, cohort retention, customer lifetime value, shipping lead times, product profitability and forecast models.

Submission Files

Submit this PDF together with the original Power BI .pbix file. The PBIX retains the editable report pages, visuals, data model and measures.